

SAPFULL_2024

SAP journal-entry extract — risk assessment, AI-assisted triage and auditor review workpaper

ENGAGEMENT SUMMARY

PERIOD UNDER REVIEW	2024-12-31
SOURCE SYSTEM / EXTRACT	SAP · extract.csv (73b90d7edec63525...)
POPULATION TESTED	65,929 lines
MATERIALITY	overall 5,000,000 MAD · performance 3,500,000 MAD
FLAGGED FOR REVIEW	21,094 documents → universe of 40 (35 inspect · 5 accepted · 0 override)
BENFORD CONFORMITY (C2, INFORMATIONAL)	MAD 0.0158 — nonconformity
AUDITOR	jdoe · decisions hash-chained and verified
REPORT STATUS	FINALIZED — All finalize gates passed (review completeness, procedure completeness, citation validity, limitation acceptance).

Audit Report — SAPFULL_2024

generated 2026-08-25 02:59 · **FINALIZED — gates passed**

65,929 POPULATION LINES	21,094 FLAGGED DOCUMENTS	40 REVIEW UNIVERSE	35 INSPECT	5 ACCEPTED	0 OVERRIDES
chain verified DECISION LOG INTEGRITY			7 LLM TURNS LOGGED		

EXECUTIVE ASSESSMENT

We tested 65,929 journal lines for the period ending 2024-12-31; 21,094 documents flagged by ten risk rules were ranked into a 40-item review universe. The reviewer directed 35 item(s) for substantive inspection and accepted 5 as adequately supported. The model rated 1 item(s) high-concern; each received a human decision. Currency coverage gaps restricted ranking to the base currency; the excluded populations are documented (Section 3) and constitute a scope limitation requiring acknowledgment. 3 document(s) failed the balance check (debits ≠ credits) — impossible without error or manipulation; each is individually decided below.

FINDINGS & OBSERVATIONS

Risk-rated, 5C structure (criteria · condition · cause · consequence · corrective action). Highest risk first.

HIGH Journal documents that do not balance	
CRITERIA	Every journal document must net to zero across its lines (double-entry principle).
CONDITION	3 document(s) in the population fail the balance check within tolerance.
CAUSE	Possible causes include extract truncation, manual keying error, or deliberate manipulation of line amounts.
CONSEQUENCE	Unbalanced postings indicate a breakdown of the fundamental control that makes journal analysis reliable; if present in the ledger they misstate account balances directly.
CORRECTIVE ACTION	Vouch each listed document to its source and ledger image; if the ledger agrees, escalate as a potential override of controls and consider fraud-response obligations (ISA 240 §32).

MEDIUM Manual-entry concentration by flagged user H.AITLA	
CRITERIA	High-risk users (per engagement risk assessment) should generate few manual entries, each independently supported.
CONDITION	20 of 40 reviewed documents were posted manually by H.AITLA, including period-end asset acquisitions routed through an account pair absent from the system baseline.
CAUSE	Asset acquisitions posted manually outside the standard procurement workflow suggest either an approval bypass or an unconfigured automated interface.
CONSEQUENCE	Concentration of manual authority in one flagged user over large, just-below-threshold acquisitions weakens segregation of duties and raises structuring risk.
CORRECTIVE ACTION	Enquire on capex approval workflow; re-assign posting rights; require dual authorization above a defined threshold; perform collective evaluation of the series.

LOW First-digit distribution deviates from Benford expectation	
CRITERIA	Benford's Law is an informational analytical (amendment C2); journal populations are not naturally Benford-distributed.
CONDITION	MAD 0.0158 — nonconformity on Nigrini bands.
CAUSE	Round-amount conventions (MAD bookkeeping, 10k multiples), threshold clustering, or genuine digit-pattern anomalies.
CONSEQUENCE	None by itself; deviation informs where to point inquiry, never concludes fraud.
CORRECTIVE ACTION	Use the digit-level chart (Section 2) to target inquiry; correlate spikes with round-amount and splitting rules.

1 • DETERMINISTIC RULE RESULTS

Ten rules executed in canonical order on the frozen extract. Recall-first design: every hit enters cross-reference ranking; precision is restored by human review. Expand a rule for its distribution and top hits.

RULE	WHAT IT DETECTS	FLAGS	% OF POP.	OUTCOME
Manual entries flags_manual_entries	Lines posted manually (mapped type or derived human username) — the primary journal-entry fraud vehicle.	56,012	85.0%	ranked
Period-end proximity flags_period_end	Manual entries within the configured pre/post-close window around period end, including post-close postings.	1,184	1.8%	review driver
Round amounts flags_round_amounts	Large amounts divisible by the round-number multiple above the materiality floor.	52	0.1%	review driver
Date divergence flags_date_divergence	Document date vs posting date gaps and entries created after period end but posted in-period (backdating signal).	16	0.0%	review driver

Entry splitting flags_entry_splitting	Clusters of manual lines just below the split threshold to one account within a short window (salami tactics).	920	1.4%	review driver
Balance check flags_balance_check	Documents whose debit/credit lines fail to net to zero within tolerance — impossible without error or manipulation.	3	0.0%	review driver
Unusual users flags_unusual_users	Rare manual users and users on the configured high-risk list.	22,325	33.9%	ranked
Unusual account pairs flags_unusual_pairs	Debit/credit account pairs absent from the system-entry baseline (a document never baselines itself).	22,632	34.3%	ranked
Reversals flags_reversals	Near-exact negations of in-period entries shortly after period end (profit-and-loss fix pattern).	0	0.0%	clean
High-risk system pairs flags_high_risk_system_pairs	Informational screen — system documents by high-risk users on accounts unusual for that user. Never gates.	0	0.0%	clean

Manual entries

56,012 flagged

Period-end proximity

1,184 flagged

Round amounts

52 flagged

– Date divergence

16 flagged

Document date vs posting date gaps and entries created after period end but posted in-period (backdating signal).

Month	Value
24-01	Low
24-05	Low
24-08	Medium
24-10	Medium
24-11	High
24-12	Medium-Low

Monthly absolute-value profile of flagged lines.

TOP ENTRIES BY AMOUNT	DATE	USER	AMOUNT
1000001902 L1	2024-11-18	M.BENNANI	61,386 MAD
1000001902 L2	2024-11-18	M.BENNANI	-61,386 MAD
1000001904 L1	2024-11-23	K.MANSOURI	58,739 MAD
1000001904 L2	2024-11-23	K.MANSOURI	-58,739 MAD
1000001901 L2	2024-10-20	K.MANSOURI	-56,866 MAD

Entry splitting

920 flagged

— Balance check

3 flagged

Documents whose debit/credit lines fail to net to zero within tolerance — impossible without error or manipulation.

TOP ENTRIES BY AMOUNT	DATE	USER	AMOUNT
1000001919 L1		M.BENNANI	3,163 MAD
1000001920 L1		S.ELFASI	2,566 MAD
1000001921 L1		S.ELFASI	2,263 MAD

Unusual users

22,325 flagged

Unusual account pairs

22,632 flagged

Reversals

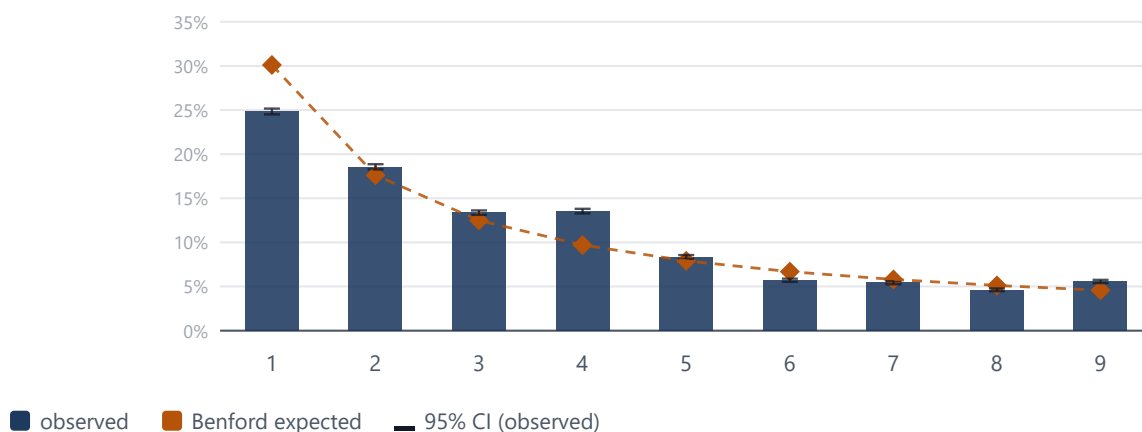
0 flagged

High-risk system pairs

0 flagged

2 · BENFORD FIRST-DIGIT CONFORMITY

Informational analytical (amendment C2): observed leading-digit distribution of all absolute amounts ≥ 1 against the Benford expectation, with 95% confidence whiskers on each observation. Informs inquiry — never gates.



MEAN ABSOLUTE DEVIATION

0.0158

Nigrini bands: **nonconformity**

Read: bars are the observed share of amounts starting with each digit; amber diamonds join the Benford curve $\log_{10}(1 + 1/d)$; whiskers are 95% confidence intervals. Whiskers crossing the diamonds are consistent with expectation.

Population analysed: **65,929** amounts. Journal populations are not naturally Benford-distributed (C2); deviations inform inquiry and never conclude fraud.

3 · REVIEW-UNIVERSE SELECTION

Workload control W1 with currency-stratified fallback X2/Y8; exclusions are limitations requiring acceptance (gate 4).

21,094 flagged docs → **40** selected for review → **40** decided → **0** pending →

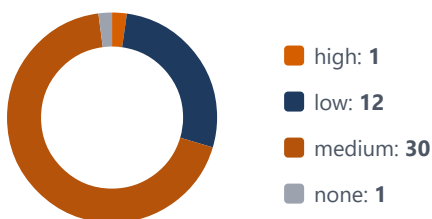
2 currencies excluded

No usable fx coverage — ranking restricted to base currency; excluded populations documented below:

CURRENCY	EXCLUDED ENTRIES	VOLUME SHARE	LARGEST EXCLUDED ENTRY
MAD	20,444	94.6%	2,313,466 MAD
EUR	610	5.4%	1,051,901 MAD

4 · LLM TRIAGE — MACHINE REASONING

Advisory assessments under rubric v1.0.0. Violet badges mark model output; the auditor table that follows is authoritative.



3000000233 **high** P4

High-risk user H.AITLA, manual entry near period end, unusual account pair 23320000 / 44110000 with 4 rules hit.

1900001082 **medium** P3

Configured high-risk user K.MANSOURI and manual entry with 2 rules hit. Multi-line invoice posting.

1900001168 **medium** P3

Entry posted by configured high-risk user H.AITLA and classified as a manual entry (derived). Multiple risk characteristics present.

remaining 37 assessed documents

5 · AUDITOR REVIEW — HUMAN DECISIONS

Substantive judgments recorded hash-chained (supersessions preserve history). This table is the authoritative record of the engagement.

ENTRY	DECISION	AUDITOR REASONING
1900000784	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900000976	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900001082	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900001168	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900001217	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900001466	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900001889	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900002902	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900003043	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900003110	accept	Routine opex, system-generated (SAP IDoc). Classifier note: IDOC_AUTO matches no configured system-user pattern so is derived manual - configuration observation raised. Vouching sample only. ox-alpha (auditor) · 2026-08-24T13:52
1900003647	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900004032	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52

1900004187	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900005065	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900005190	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
1900006879	accept	Routine opex, system-generated (SAP IDoc). Classifier note: IDOC_AUTO matches no configured system-user pattern so is derived manual - configuration observation raised. Vouching sample only. ox-alpha (auditor) · 2026-08-24T13:52
1900006888	accept	Routine opex, system-generated (SAP IDoc). Classifier note: IDOC_AUTO matches no configured system-user pattern so is derived manual - configuration observation raised. Vouching sample only. ox-alpha (auditor) · 2026-08-24T13:52
1900007250	accept	Routine opex, system-generated (SAP IDoc). Classifier note: IDOC_AUTO matches no configured system-user pattern so is derived manual - configuration observation raised. Vouching sample only. ox-alpha (auditor) · 2026-08-24T13:52
1900007404	accept	Routine opex, system-generated (SAP IDoc). Classifier note: IDOC_AUTO matches no configured system-user pattern so is derived manual - configuration observation raised. Vouching sample only. ox-alpha (auditor) · 2026-08-24T13:52
1900007643	inspect	Material or undocumented item outside routine patterns - individual substantive testing required. ox-alpha (auditor) · 2026-08-24T13:52
3000000013	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000034	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000112	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000182	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000212	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52

3000000233	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000242	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000244	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000348	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000360	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000364	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000429	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000436	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000459	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000482	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000556	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000591	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52

3000000597	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000675	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52
3000000712	inspect	Fixed-asset acquisition via unusual pair 23320000/44110000 by high-risk user H.AITLA; ~2M MAD series pattern. Vouch to supplier invoice/PO/delivery note; confirm FA register posting; collective series evaluation applies. ox-alpha (auditor) · 2026-08-24T13:52

6 · WORKPAPER NARRATIVE

Drafted by the model strictly from the keyed facts block; citations resolved inline (gate 3 validates every reference).

Scope and Approach

For audit run SAPFULL_2024 covering the period ending **2024-12-31**, we tested a population of **65929** journal lines. A total of **21094** documents met initial flagging criteria from **21094** total flagged items before capping. The review universe was established at **40** items using **currency-stratified fallback**.

Flagging Results

Risk rules generated the following hit distributions across the dataset: **56012** manual entries, **22632** unusual pairs, **22325** unusual users, **1184** period-end items, **920** entry splitting instances, **16** date divergence instances, **52** round amounts, **3** balance check failures, **0** reversals, and **0** high-risk system pairs.

Findings and Observations

Testing identified 3 balance check failures (**3**), indicating instances where debits do not equal credits within tolerance. Additionally, manual-entry concentration by high-risk user H.AITLA and first-digit Benford distribution deviations were observed across the flagged population.

Review Outcomes

Out of the reviewed items, the reviewer directed **35** items for substantive inspection and accepted **5** items as adequately supported, with **0** override decisions recorded.

Limitations and Follow-ups

Currency coverage gaps restricted ranking to the base currency for run ID **40**, representing a scope limitation that requires appropriate acknowledgment and follow-up procedures in subsequent audit phases.

Mandatory limitations (§11)

- 1. Vouching is external: entries marked 'inspect' leave this tool for substantive testing against supporting documentation; this workpaper lists them, it does not vouch them.
- 2. Benford analysis is informational: journal amount populations are not naturally Benford-distributed; results inform inquiry, never conclude.
- 3. System-entry account pairs are not screened: unusual_pairs defines its baseline from system entries of the current period, excluding previously flagged documents.
- 4. Reviewer identity is declared, not authenticated, until SSO (reviewer_source distinguishes); decision rows are hash-chained so tampering is detectable, but identity is not yet non-repudiable.
- 5. Statistical outlier results are seeded and recorded; seeds live in the run store.
- 6. LLM outputs are documented, not deterministic — the reproducibility guarantee covers the deterministic path; the judgment path is fully logged instead.
- 7. Triage follow-ups are recorded, not executed — acting on one is a deliberate new run.
- 8. The representative sample supports targeted coverage, not statistical projection.
- 9. LLM-facing free text is sanitized, not guaranteed safe: descriptions are delimited untrusted data and scanned for injection patterns; prompt-level defense is a mitigation, and the structural backstop is schema-constrained advisory output behind human review.
- 10. PII scrubbing is pattern-based: regex classes and configured redaction terms remove structured identifiers from LLM-bound text but cannot catch all sensitive prose.

DQ acknowledgments & integrity

WARNING	REASON	REVIEWER
none acknowledged this run		
deterministic engine	LLM triage + narrative	human decisions
hash chain VERIFIED		

Scope & use. This tool is a risk-flagging, prioritization, documentation and review-workflow system. It is not a substitute for substantive testing or vouching, not a complete fraud detection system, and provides no assurance that no material misstatement exists. Entries marked “inspect” leave this tool for substantive testing against supporting documentation; this report lists them — it does not vouch them. LLM outputs are documented, not deterministic; the deterministic path is reproducible and the judgment path fully logged and tamper-evident.